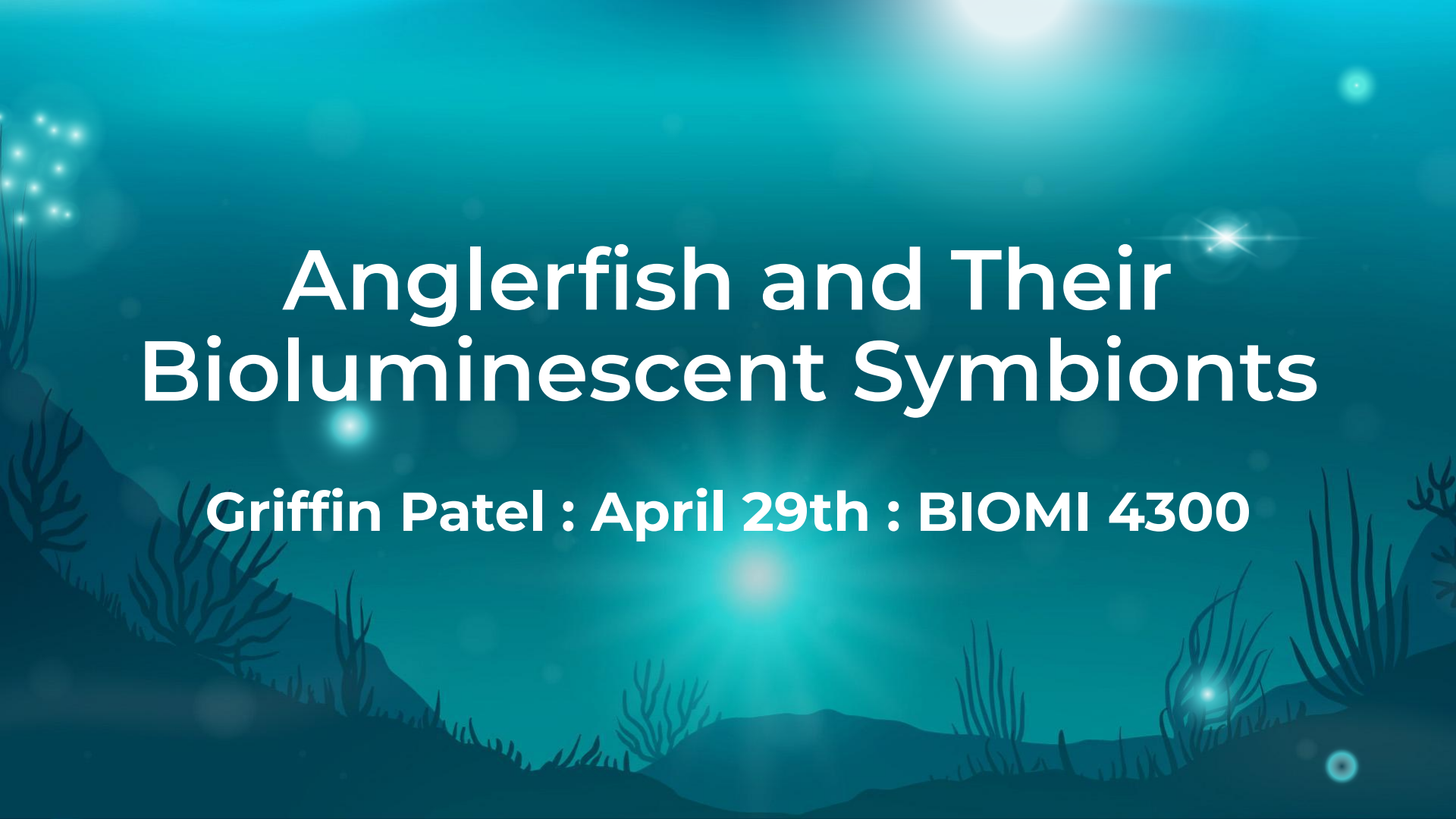


The background is a deep teal color with a gradient, darker at the bottom and lighter at the top. There are several bright, out-of-focus light spots scattered throughout, some with lens flare effects. Silhouettes of coral and seaweed are visible along the bottom edge, and a faint, glowing light source is visible in the upper right corner.

Anglerfish and Their Bioluminescent Symbionts

Griffin Patel : April 29th : BIOMI 4300

Roadmap

1. Background and Context
2. Overarching Scientific Question
3. Questions and hypothesis
 - a. Biodiversity
 - b. Composition
 - c. Taxa-specific
4. Conclusion and Takeaways
5. Future Experiments



[CNN](#)

Ceratioid Anglerfishes of the Gulf of Mexico

- A suborder of Lophiiformes, the main classification of anglerfish
- All samples are from female specimens due to sexual dimorphism
- Main light producing organ, the **Esca**
 - Other auxiliary light organs depending on species
 - ie. **caruncles** (only in Ceratiidae)



Where
caruncles
would be
located

*Esca - Lure Tip
(bioluminescent bulb)*

*Illicium - Lure Rod
(modified dorsal
spine extending from
the forehead)*

*Dorsal Fin
(provides stability)*

*Caudal Fin
(main fin used
for propulsion)*

Small eyes

*Mouth
(wide and extendable -
capable of swallowing
prey larger than itself)*

*Pectoral Fin
(for steering)*

*Pelvin Fin
(used for
balance)*



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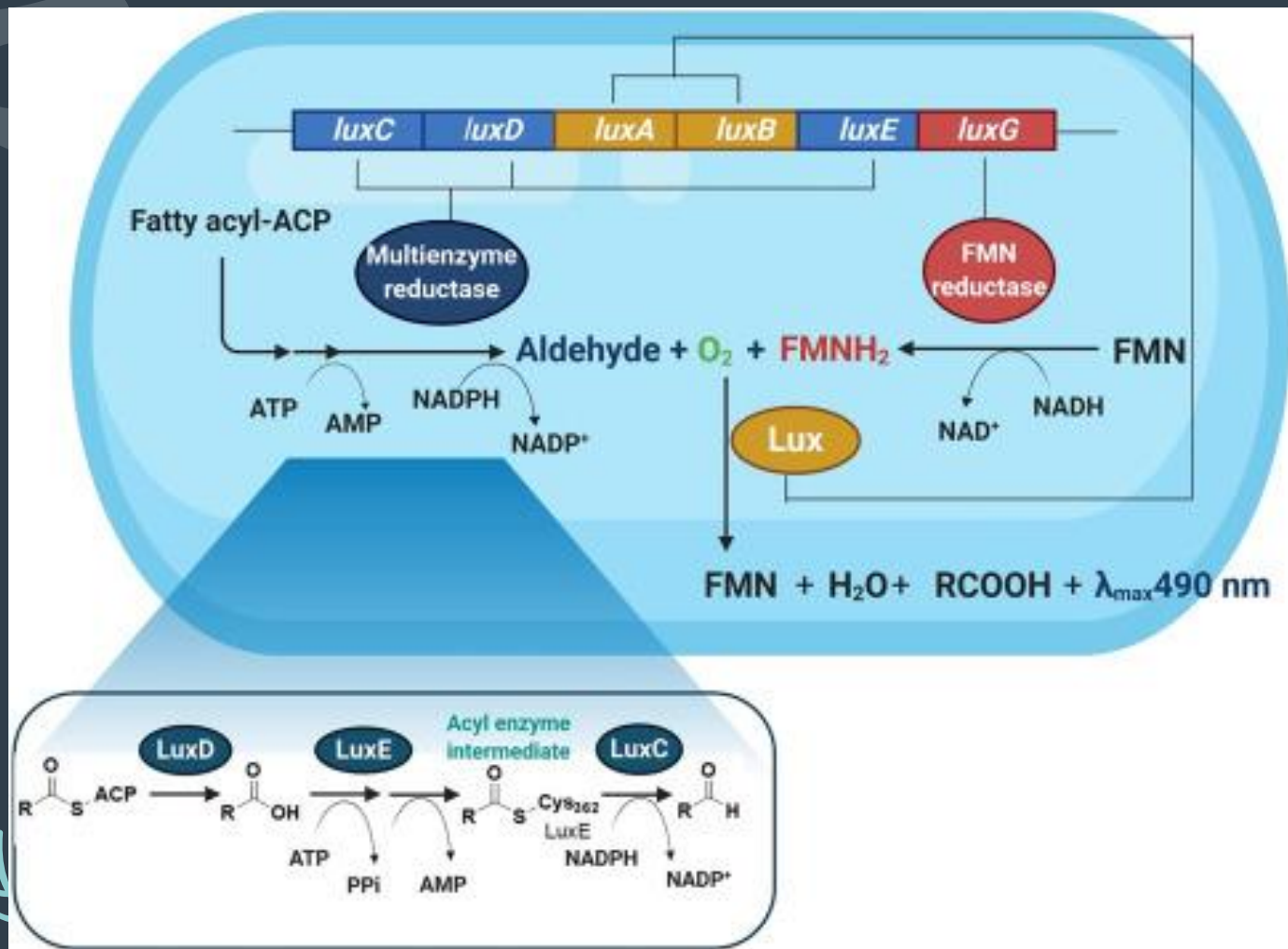
Natura

How Is Anglerfish Bioluminescence Achieved?



Bacterially mediated luminescence

- The light in ceratioid anglerfish is produced by **specialized *Enterovibrio* symbionts** in the esca and caruncles
- *Enterovibrio luxaltus* and *Enterovibrio escacola* are the bacteria responsible for bioluminescence in my
- sampled ceratoides [Hendry et al. 2018](#)
- Light is produced via the **Lux Operon**
 - Quorum sensing regulated expression of the enzyme luciferase that reduces FMN, oxygen, and a long-chain aldehyde to produce light [Science direct](#)



Do luminous and non-luminous anglerfish tissues differ in their microbial community composition?

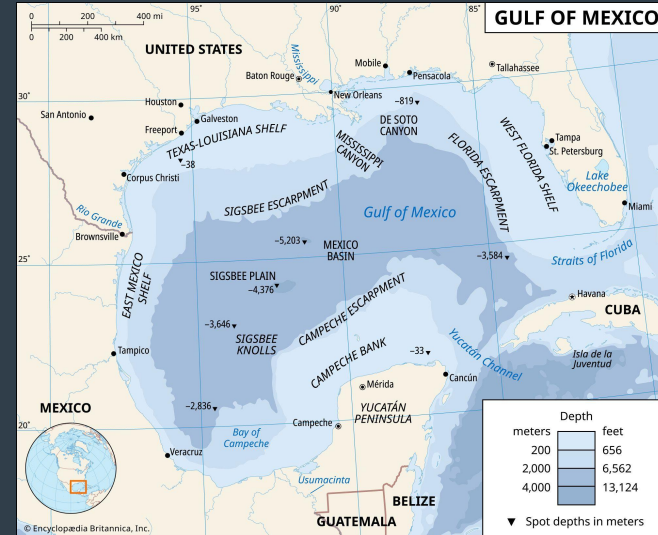
- Simple question on the surface, but:
- Are these symbionts present everywhere and only expressing Lux proteins in light organs?
- What are the overall differences in biodiversity between luminous and non-luminous tissues?
- Are light producing bacteria having effects on the other bacteria present?

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- Simple question on the surface, but:
 - Are these symbionts present everywhere and only expressing Lux proteins in light organs?
 - What are the overall differences in biodiversity between luminous and non-luminous tissues?
 - Are light producing bacteria having effects on the other bacteria present?

Data about Data: How samples were collected and processed

- All anglerfish were collected over the course of four DEEPEND cruises aboard the R/V Point Sur in the Gulf of Mexico
- Luminous tissues: Esca, Caruncles
- Non-luminous tissues: fin, gills, guts, illicium, skin
- 16S rRNA V4 region with 515F / 806R primers
- 2x250 bp Illumina MiSeq sequencing
- ASVs assigned with DADA2
- Final sequencing-depth range: 914–1,034 reads
- Community structure analyzed with Sorensen, Bray-Curtis, PERMANOVA, and dispersion tests



Luminous anglerfish tissues are predicted to host distinct and more specialized microbial communities

Part 1 - Biodiversity: How does microbial biodiversity differ between luminous and non-luminous anglerfish tissues?

Part 2 - Composition: How does microbial community composition differ between luminous and non-luminous tissues?

Part 3 - Taxa-specific responses: Which taxonomic groups are most associated with luminous tissues?



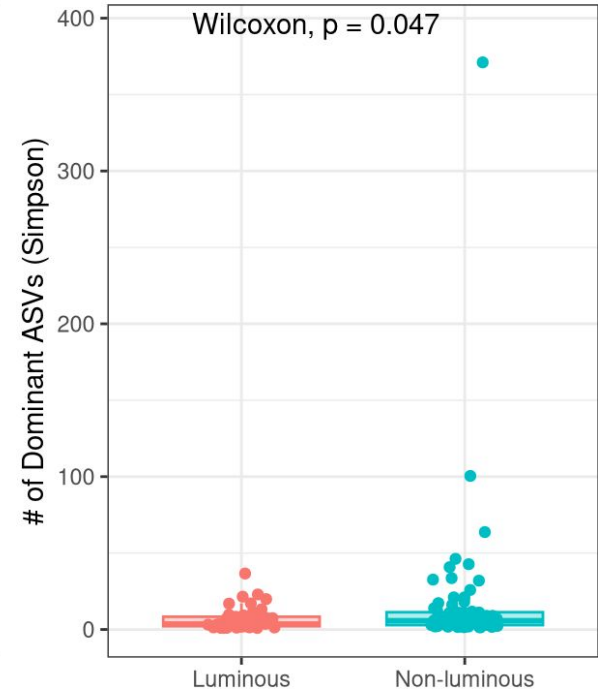
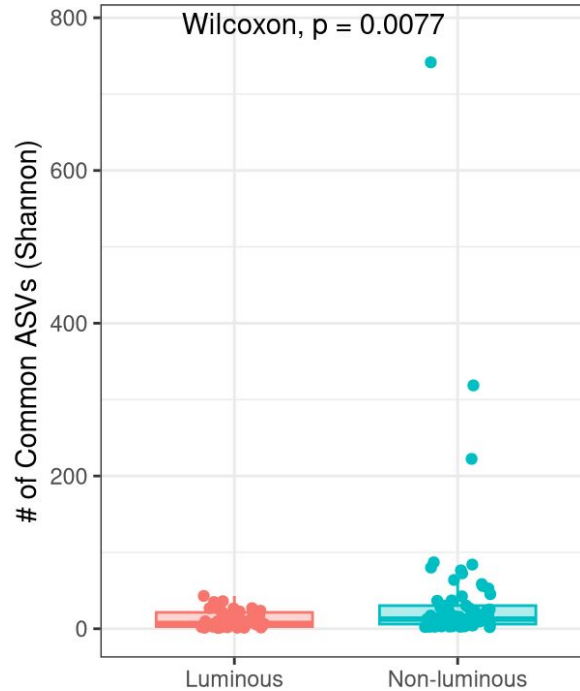
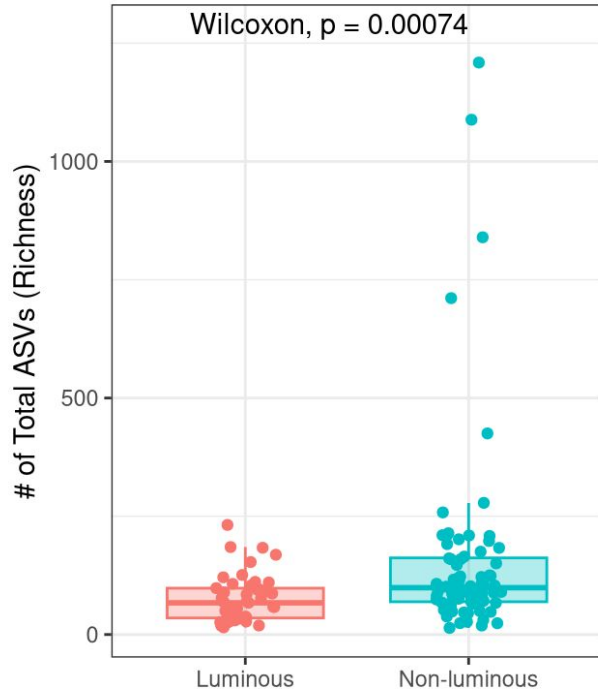
Microbial communities are predicted to differ between luminous and non-luminous anglerfish tissues

Part 1 - Biodiversity: How does microbial biodiversity differ between luminous and non-luminous anglerfish tissues?

- Hypothesis: Luminous tissues will show different biodiversity patterns than non-luminous tissues because they may be enriched for specialized symbiont-associated bacteria.
- The specialized nature of light organs provides a unique environment that should have a distinct community



Non-Luminous Tissues Consistently have Higher Diversity



Luminous tissues appear more specialized and less biodiverse

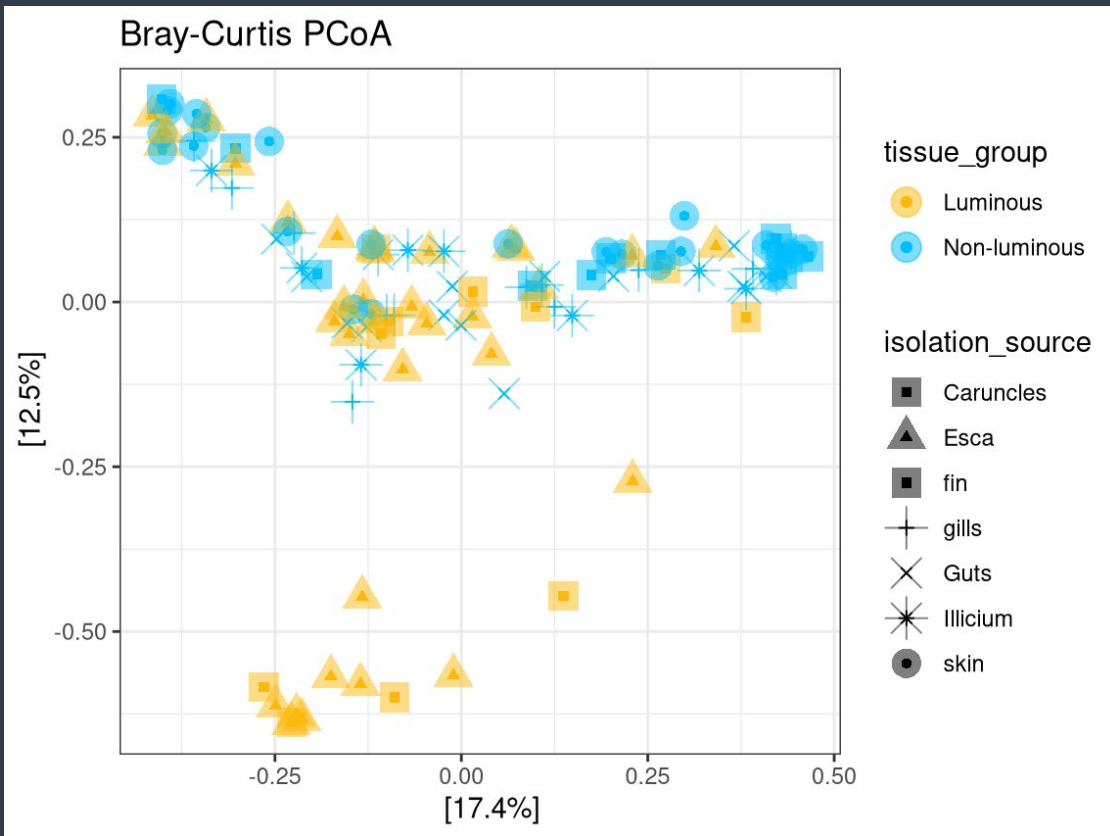
- Non-luminous tissues have higher richness
 - They contain more total ASVs than luminous tissues.
- Non-luminous tissues are also more diverse when abundance is considered
- Shannon and Simpson diversity remain higher outside the luminous organs.
 - Luminous tissues appear more specialized
 - This pattern is consistent with enrichment of a narrower set of bacteria, likely including the endosymbionts.

Microbial communities are predicted to differ between luminous and non-luminous anglerfish tissues

Part 2 - Composition: How does microbial community composition differ between luminous and non-luminous tissues?

- Hypothesis: There will be unique compositions between tissue groups, especially when abundance is considered.
 - Luminous tissues should contain a distinct microbial signature associated with bioluminescent symbionts

Luminous Tissues Demonstrate Unique Communities



Permanova Results:

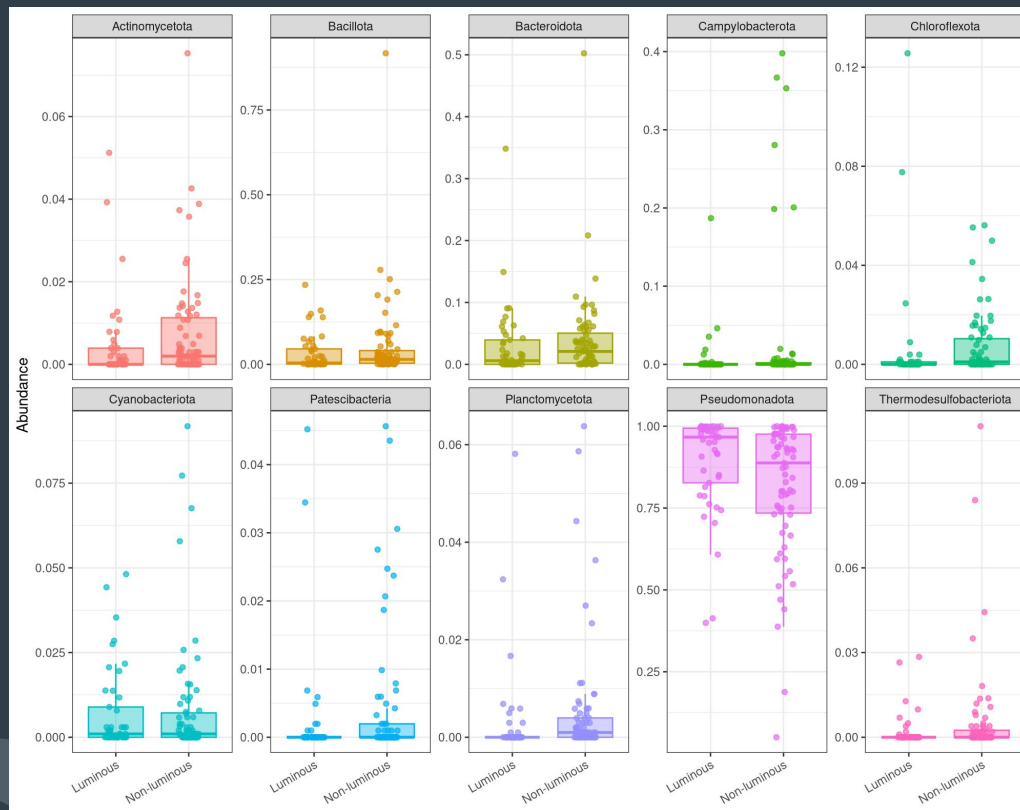
- Df = 1
 - Tissue Group
- $R^2 = 0.0432$
 - 4.32% of variation explained
 - Not much, but significant
 - $p = 0.001$
- $F = 5.1477$
 - Strong between-group differences

Microbial communities are predicted to differ between luminous and non-luminous anglerfish tissues

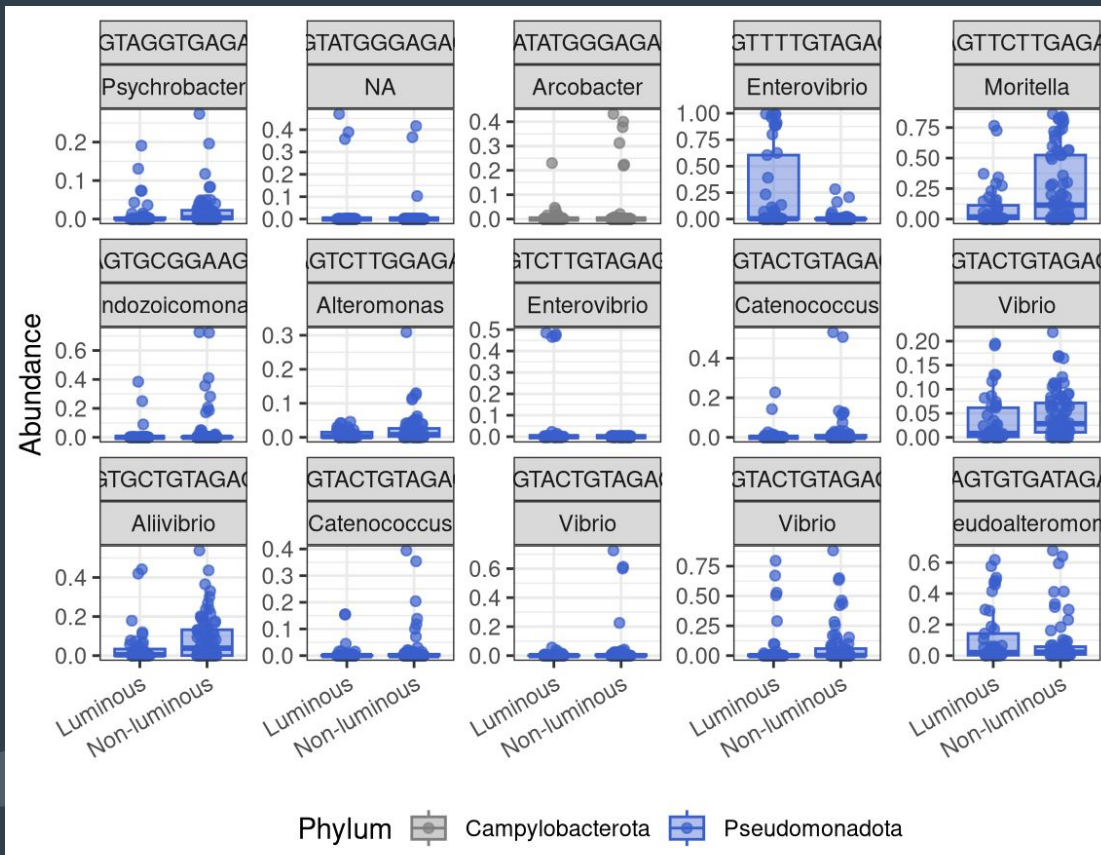
Part 3 - Taxa-specific responses: Which taxonomic groups are most associated with luminous tissues?

- Hypothesis: Specific bacterial taxa, with a **heavy** focus on *Enterovibrio*, should be enriched in luminous tissues, reflecting the specialized communities in these organs
 - *Enterovibrio* are expected to be present in other tissues as well, to lesser degrees.

Pseudomonadota dominates microbial communities across all tissue types



14 of the Top 15 ASVs are Pseudomonadota



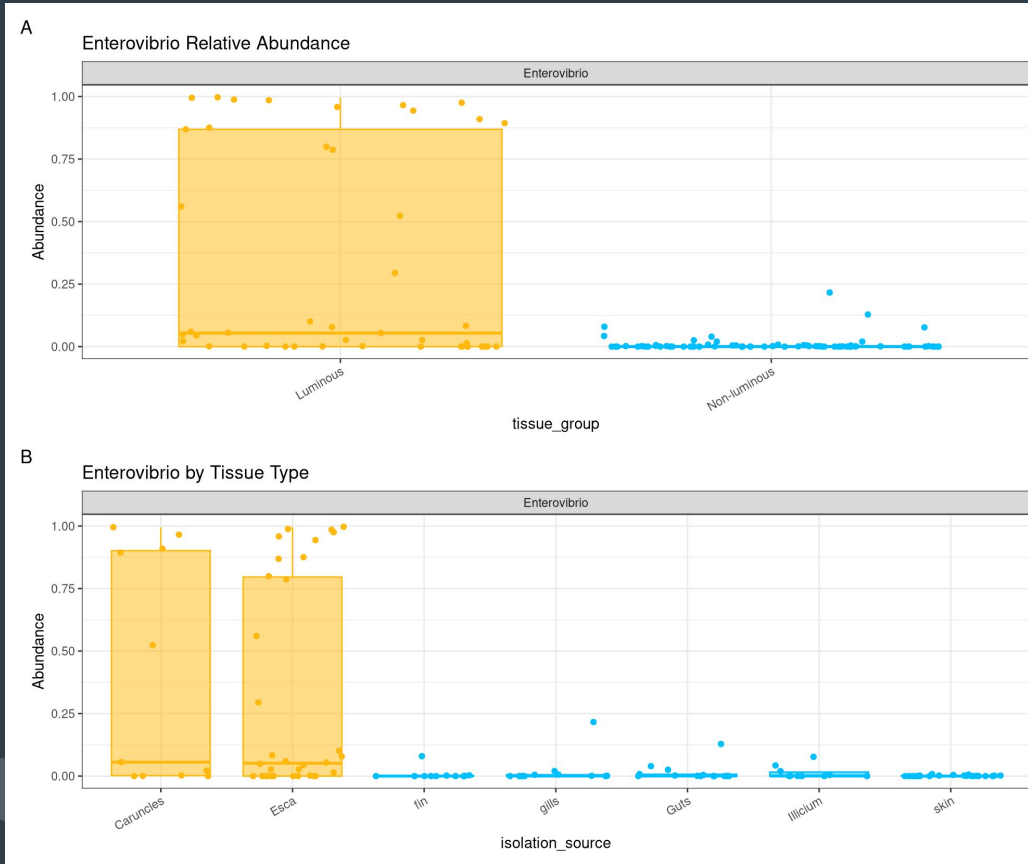


Pseudomonadota, formerly Proteobacteria, are Dominant in the Ocean in general

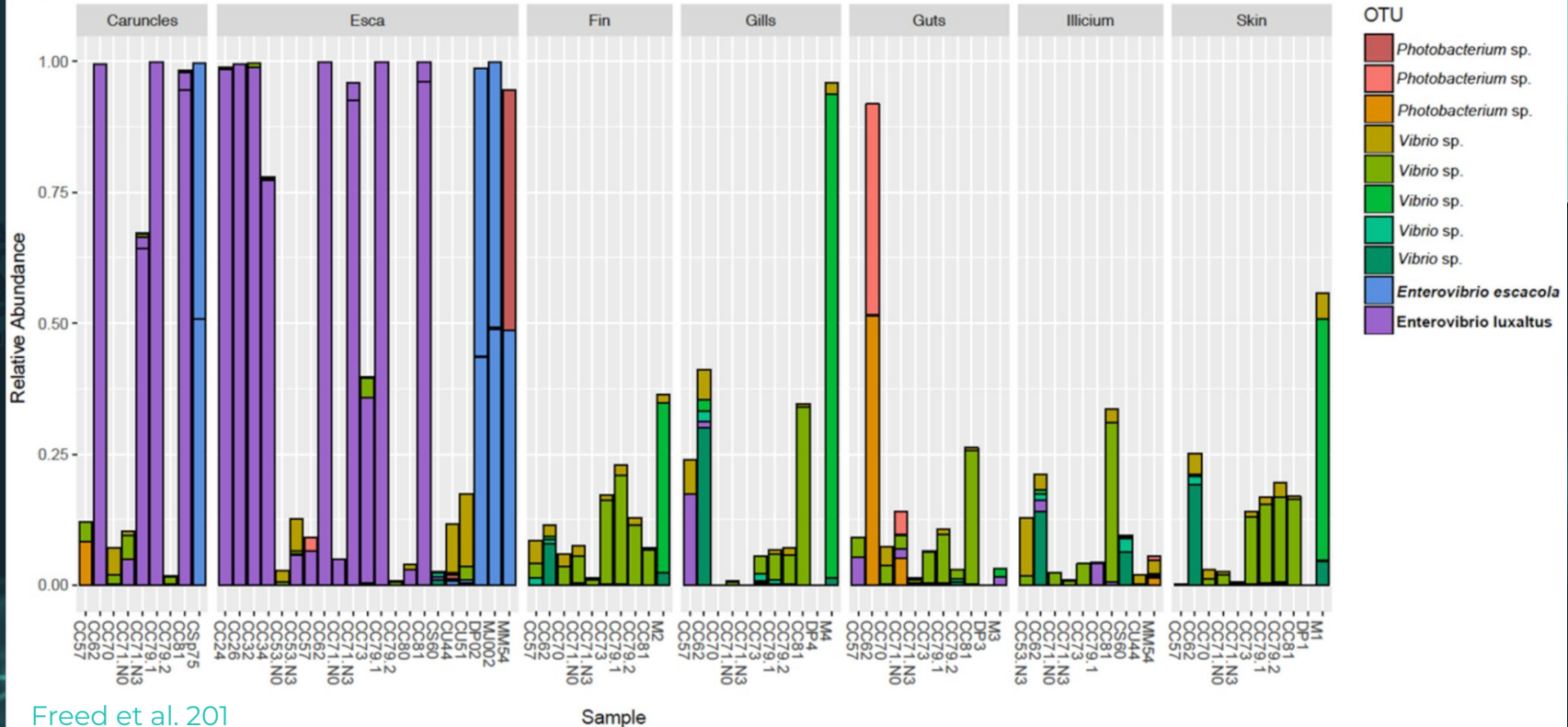
- The ocean has many diverse environments
- Pseudomonadota are highly adaptive and thrive in many marine microbial communities [Zheng et al 2022](#)
- The deep ocean is a more specialized environment, **still dominated by pseudomonadota**
- So not shocking results alone, but within the phyla, **enriched enterovibrio** in luminous tissues is
- Further support for higher diversity in non-luminous tissues



Enterovibrio Are Heavily Enriched In Light Organs



(A)



Luminous anglerfish tissues are associated with a distinct, abundance-weighted microbial signal consistent with symbiont enrichment

- Anglerfish support **highly specialized microbial communities**
- Enterovibrio was the clearest taxa linked to the luminous tissues
- Congruent results with the source paper
- **Overall, luminous tissues appear more specialized, while non-luminous tissues support broader microbial diversity.**



[inaturalist](#)



Future Experiments



- Comparing seawater populations of enterovibrio to anglerfish populations
- Genetic conservation of other bacteria in light-organs
- Phylogenetic trees of luminous and non-luminous tissues



Questions?

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